

Bayes Theorem -HW

1.

An individual has 3 different email accounts. Most of her messages, in fact 70%, come into account #1, whereas 20% come into account #2 and the remaining 10% into account #3. Of the messages into account #1, only 1% are spam, whereas the corresponding percentages for accounts #2 and #3 are 2% and 5%, respectively. What is the probability that a randomly selected message is spam?

2.

A total of 600 of the 1,000 people in a retirement community classify themselves as Republicans, while the others classify themselves as Democrats. In a local election in which everyone voted, 60 Republicans voted for the Democratic candidate, and 50 Democrats voted for the Republican candidate. If a randomly chosen community member voted for the Republican, what is the probability that she or he is a Democrat?

3.

A medical centre encourages elderly people to have a flu vaccination each year. The vaccination reduces the likelihood of getting flu from 40% to 10%. If 45% of the elderly people visiting the centre have the vaccination, find the probability that an elderly person chosen at random

a) gets flu b) had the vaccination, given that they get flu.

4.

A shopkeeper buys one third of his stock of light bulbs from Company X, and the rest from Company Y.

An independent report states that 3% of light bulbs from Company X are faulty and that 2% from Company Y are faulty.

- a) If the shopkeeper chooses a bulb at random from his stock and tests it, what is the probability that it is faulty?**
- b) If the bulb is faulty, what is the probability that it came from Company Y?**

5.

Two firms V and W consider bidding on a road-building job, which may or may not be awarded depending on the amounts of the bids. Firm V submits a bid and the probability is 0.8 that it will get the job provided firm W does not bid. The probability is 0.7 that W will bid, and if it does, the probability that V will get the job is only 0.4.

- (a) What is the probability that V will get the job?**
- (b) If V gets the job, what is the probability that W did not bid?**

6.

A large firm has 85% of its service calls made by a contractor, and 10% of these calls result in customer complaints. The other 15% of the service calls are made by their own employees, and these calls have a 5% complaint rate. Find the

- (a) probability of receiving a complaint.
- (b) probability that the complaint was from a customer serviced by the contractor.